MINERVA CryoCell P&ID — Colour Scheme Revision Guide (v5)

Project: MINERVA CryoCell — SCK CEN · MYRRHA / MINERVA Phase 1
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Status: S2 — FOR ACCEPTANCE · **RESTRICTED**
Standards: SCK CEN AD_01.16 · ISO 10628 · ANSI/ISA-5.1-2022 · IEC 60617

1. Purpose of the revision

The v5 revision replaces the legacy colour assignment (which used red for the 40 K shield and treated all helium circuits with broadly similar emphasis) with a **cryogenic-focus palette**. The guiding principle is that **colour now encodes temperature**: the coldest lines read coolest (deep blue / cyan), the thermal shield reads warm (orange / red) and the warm piping system reads green / olive. This makes the thermal hierarchy of the cryomodule legible at a glance, on A3 and after monochrome plotting.

2. New palette

Lines are grouped into three temperature zones. Each **primary** (main) line has a saturated hue; its **branch** lines (designator with a prime, e.g. `A′`) use a darker or lighter shade of the same hue.

Cold header (top of sheet)

Line	Designator	Colour	Hex	Service	Temp	Pressure	Flow	DN	MOC
A	A	Blue	`#0000FF`	4.5 K primary helium	4.5 K	3 bar	~50 g/s	DN50	SS316L
A′	A′	Navy	`#000080`	4.5 K branches	4.5 K	3 bar	varies	DN25	SS316L
B	B	Cyan	`#00FFFF`	2 K primary helium	2 K	27 mbar	~47.5 g/s	DN40	SS316L
B′	B′	Teal	`#008B8B`	2 K branches	2 K	27 mbar	varies	DN25	SS316L

Thermal shield

Line	Designator	Colour	Hex	Service	Temp	Pressure	Flow	DN	MOC
D	D	Orange	`#FF8000`	40 K shield inlet	40 K	14 bar	TBD	DN32	Cu
D′	D′	Light orange	`#FFB366`	40 K branches	40 K	14 bar	varies	DN20	Cu
E	E	Red	`#FF0000`	60 K shield outlet	60 K	13 bar	TBD	DN32	Cu
E′	E′	Dark red	`#CC0000`	60 K branches	60 K	13 bar	varies	DN20	Cu

Warm piping system (WPS) — bottom of sheet

Line	Designator	Colour	Hex	Service	Temp	Pressure	Flow	DN	MOC
W	W	Green	`#00FF00`	WPS warm return	4.5 K → 300 K	6 bar	~2.5 g/s	DN20	SS304
S	S	Lime	`#BFFF00`	WPS service / safety	2 → 292 K	1.05 bar	TBD	DN15	SS304
U	U	Olive	`#808000`	WPS GHe supply inlet	292 K	14 bar	TBD	DN15	SS304

Reference

Item	Colour	Hex	Notes
Outside scope	Grey (dashed)	`#808080`	Services beyond the hand-over boundary
Secondary / DI-water	Green-blue	per sheet	Cooling water utility

3. Old → new mapping

Element	v4 (old)	v5 (new)	Rationale
4.5 K primary (A)	blue `#0000FF`	**blue** **`#0000FF`**	retained — coldest reads coolest
2 K primary (B)	cyan `#00FFFF`	**cyan** **`#00FFFF`**	retained
40 K shield in (D)	red `#FF0000`	**orange** **`#FF8000`**	red re-assigned to the outlet; 40 K reads warm-orange
60 K shield out (E)	(grouped / olive)	**red** **`#FF0000`**	hottest cryo line gets the hottest hue
Warm return (W)	green `#00FF00`	**green** **`#00FF00`**	retained, now anchors the WPS zone
Service / safety (S)	—	**lime** **`#BFFF00`**	new explicit WPS line
GHe supply (U)	olive `#808000`	**olive** **`#808000`**	retained

The key behavioural change: **red moved from the 40 K inlet to the 60 K outlet**, and the 40 K inlet became orange — so the shield pair D→E now reads as a clear warm-up (orange → red) instead of two reds.

4. Branch (prime) shading rule

- Cold lines (A, B): branch = **darker** shade (`A′` navy, `B′` teal).
- Thermal lines (D, E): branch = **lighter / darker** shade of the same hue (`D′` light orange, `E′` dark red).
- A run is classified **primary** vs **branch** automatically using the 55th-percentile run-length threshold per class; primaries live on the main piping layers (04/05/06) and branches on the `_BRANCHES` layers (04B/05B/06B).

5. Where the scheme is applied

- **Production sheets** — all 16 (QCELL + RFCELL × 2 sheets × STANDARD / CONTROL-CENTRIC × colour / mono).
- **Zone bands** (layer `02C_Zone_Bands`) tint the sheet into cold-header, thermal-shield, equipment and warm-WPS reading zones.
- **Line labels** (layer `04D_Piping_LINE_LABELS`) carry `[LINE]-[SIZE]-[MOC]`.
- **Flow arrows** (layer `04G_Flow_Arrows`) are coloured per line.
- **Legend** — the on-sheet legend now contains the full **line specification table** (colour swatch, temperature, pressure, DN, MOC).
- **MAIN-LINES-ONLY** schematic and the `VIEW_MAINLINES_ONLY` preset.

6. Monochrome behaviour

When colour is removed, the class hierarchy is carried by **line weight + dash pattern** instead of hue:

Zone	Mono weight	Dash
Cold header (A/B)	1.0 mm primary / 0.6 mm branch	solid
Thermal shield (D/E)	0.8 mm primary / 0.6 mm branch	solid
Warm WPS (W/S/U)	0.5 mm	dashed (8,3)

This guarantees the drawing survives black-and-white plotting and photocopying without loss of meaning.

7. Accessibility / plotting notes

- Blue↔cyan and orange↔red pairs were chosen to remain distinguishable by position (zone bands) even for colour-vision-deficient readers; the mono weight/dash system is the formal fallback.
- All hex values above are the controlled source-of-truth and match `generator/line_spec_data.py` and `LINE_SPECIFICATION_MASTER.xlsx`.